MTH108 JUL22 TOA Review

6 May 2023

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Question 1

Use various techniques of integration to evaluate the following integrals.

Question 1a

$$\int \frac{3x^2}{(x^2-3x+2)^2}\,dx$$

(7 marks)

Strategy:

- This is a rational function
- factorise the denominator
- split into partial fractions
- integrate the partial fractions

$$\frac{3x^2}{(x^2-3x+2)^2} = \frac{3x^2}{\underbrace{(x-1)^2(x-2)^2}_{\text{repeated root form}}} = \frac{A}{x-1} + \frac{B}{(x-1)^2} + \frac{C}{x-2} + \frac{D}{(x-2)^2}$$

Degree of numerator is 2, less than the degree (4) of the denominator $\Rightarrow$ no need to perform long division

$$3x^2 \equiv A(x-1)(x-2)^2 + B(x-2)^2 + C(x-1)^2(x-2) + D(x-1)^2$$
$$\equiv A(x-1)(x^2-4x+4) + B(x^2-4x+4) + C(x^2-2x+1)(x-2) + D(x^2-2x+1)$$
$$\equiv A(x^3-5x^2+8x-4) + B(x^2-4x+4) + C(x^3-4x^2+5x-2) + D(x^2-2x+1)$$
$$\equiv (A+C)x^3 + (B-5A-4C+D)x^2 + (8A-4B+5C-2D)x + (4B-4A-2C+D)$$

$\Rightarrow A+C=0$ —(1) $\Rightarrow 8A-4B+5C-2D=0$ —(3)

$\Rightarrow B-5A-4C+D=3$ —(2) $\Rightarrow 4B-4A-2C+D=0$ —(4)

Strategy: eliminate 1 variable at a time

from (1), $C=-A$ [eliminate C]

from (2)-(4), $(B-5A-4C+D)-(4B-4A-2C+D)=3-0$ [D cancels out]

$-3B-A-2C=3$

$-3B-A-2(-A)=3$ ← sub $C=-A$

$-3B+A=3$

$A=3+3B$ [eliminate A]

From (2), $B=3+5A+4C-D$

$=3+5A+4(-A)-D$ sub $C=-A$

$=3+A-D$

$=3+(3+3B)-D$ sub $A=3+3B$

$=6+3B-D$

$D=6+2B$ [Eliminate D]

$\therefore A=3+3B,\ B=B,\ C=-A=-3-3B,\ D=6+2B$

A, C, D are derived from (1), (2), (4), so must use (3) to derive value of B

from (3), $8A-4B+5C-2D=0$

$8(3+3B)-4B+5(-3-3B)-2(6+2B)=0$

$B-3=0$

$\therefore B=3$ $\quad\therefore C=-12$

$\therefore A=3+3B=12$ $\quad\therefore D=6+2B=12$

1(a) $\displaystyle\int \frac{3x^2}{(x^2-3x+2)^2}\,dx$

$$= \int \frac{12}{x-1} + \frac{3}{(x-1)^2} - \frac{12}{x-2} + \frac{12}{(x-2)^2}\,dx$$

$$= \int \frac{12}{x-1} - \frac{(-1)(3)}{(x-1)^2} - \frac{12}{x-2} - \frac{(-1)(12)}{(x-2)^2}\,dx$$

pattern match to $\frac{d}{dx}\frac{1}{f(x)}$

$$= 12\ln|x-1| - \frac{3}{x-1} - 12\ln|x-2| - \frac{3}{x-2} + C,$$

where C is a constant

Question 1b

$$\int_4^9 \frac{\ln x}{2\sqrt{x}}\,dx$$

(7 marks)

Strategy: integration by parts

$$\int \frac{\ln x}{2\sqrt{x}}\,dx = \int \underbrace{(\ln x)}_{u}\underbrace{\left(\frac{1}{2\sqrt{x}}\right)}_{dv}\,dx$$

↳ of LIATE Rule

$\frac{d}{dx}\ln x = du$

$$= \underbrace{(\ln x)}_{u}\underbrace{x^{\frac{1}{2}}}_{v} - \int \left(\frac{1}{x}\right) x^{\frac{1}{2}}\,dx \rightarrow \int v\,du$$

$$= x^{\frac{1}{2}}\ln x - \int x^{-\frac{1}{2}}\,dx$$

$$= x^{\frac{1}{2}}\ln x - 2\int \tfrac{1}{2}x^{-\frac{1}{2}}\,dx$$

$$= x^{\frac{1}{2}}\ln x - 2x^{\frac{1}{2}}$$

$$= x^{\frac{1}{2}}(\ln x - 2)$$

$$\int_4^9 \frac{\ln x}{2\sqrt{x}}\,dx = \left[x^{\frac{1}{2}}(\ln x - 2)\right]_4^9$$

$$= 3(\ln 9 - 2) - 2(\ln 4 - 2)$$

$$= 3(2\ln 3 - 2) - 2(2\ln 2 - 2)$$

$$= 6\ln 3 - 4\ln 2 - 2$$

Question 1c

$$\int \frac{\sin^5 x}{\sqrt{\cos x}}\,dx$$

(6 marks)

Strategy: odd power of $\sin(x) \Rightarrow$ split into $\sin^4 x \cdot \sin x$.

$$\int \frac{\sin^5 x}{\sqrt{\cos x}}\,dx = \int \frac{\sin^4 x}{\sqrt{\cos x}} \sin x \, dx$$

$$= -\int \frac{\sin^4 x}{\sqrt{\cos x}} \underbrace{(-\sin x)\,dx}_{du \Rightarrow u = \cos x}$$

$$= -\int \frac{(\sin^2 x)^2}{\sqrt{\cos x}} (-\sin x)\,dx$$

$$= -\int \frac{(1-\cos^2 x)^2}{\sqrt{\cos x}} (-\sin x)\,dx$$

$$= -\int \frac{(1-u^2)^2}{\sqrt{u}}\,du$$

$$= -\int \frac{1-2u^2+u^4}{\sqrt{u}}\,du$$

$$= -\int u^{-\frac{1}{2}} - 2u^{\frac{3}{2}} + u^{\frac{7}{2}}\,du$$

$$= -\left[2u^{\frac{1}{2}} - \frac{4}{5}u^{\frac{5}{2}} + \frac{2}{9}u^{\frac{9}{2}}\right] + C$$

$$= \frac{4}{5}u^{\frac{5}{2}} - 2u^{\frac{1}{2}} - \frac{2}{9}u^{\frac{9}{2}} + C$$

$$= \frac{4}{5}(\cos x)^{\frac{5}{2}} - 2\sqrt{\cos x} - \frac{2}{9}(\cos x)^{\frac{9}{2}} + C$$

Question 2a

Solve $\dfrac{d}{dx}\displaystyle\int_{x^e}^{e^x} \sin(t^3)\,dt$.

(6 marks)

Strategy: Use FTC2

$$\int_{x^e}^{e^x} \sin(t^3)\,dt = F(e^x) - F(x^e)$$

where: $f(t) = \sin t^3$

$F(x) = \int_a^x f(t)\,dt$

a is some constant

$$\frac{d}{dx}\int_{x^e}^{e^x} \sin(t^3)\,dt = \frac{d}{dx}\left[F(e^x) - F(x^e)\right]$$

$$= e^x f(e^x) - ex^{e-1} f(x^e)$$

$$= e^x \sin e^{3x} - ex^{e-1} \sin x^{3e}$$

FTC1: $\frac{d}{dx}\left[\int_{x^e}^{e^x} \sin t^3 \, dt\right] = \frac{d}{dx}\left[\int_a^{e^x} \sin t^3 \, dt - \int_a^{x^e} \sin t^3 \, dt\right]$

$= \frac{d}{dx}\int_a^{e^x} \sin t^3 \, dt - \frac{d}{dx}\int_a^{x^e} \sin t^3 \, dt$

$= \frac{d}{du}\int_a^{u} \sin t^3 \, dt \, \frac{du}{dx} - \frac{d}{dv}\int_a^{v} \sin t^3 \, dt \, \frac{dv}{dx}$ where $u = e^x$, $v = x^e$

$= f(u)\frac{du}{dx} - f(v)\frac{dv}{dx}$

$= f(e^x)\, e^x - f(x^e)\,(ex^{e-1})$

$= e^x \sin e^{3x} - e x^{e-1} \sin x^{3e}$

Question 2b

Showing your working clearly, determine the derivatives of the following logarithmic functions:

(i) $\log_5(7x)$

(ii) $\log_5(2020x)$

(iii) $\log_5(369x)$.

What can you observe from your answers? Using the properties of logarithms, provide an explanation for your observation.

(7 marks)

Strategy: convert $\log_b f(x)$ to $\ln f(x)$

$$\log_a b = \frac{\log_c b}{\log_c a}$$ } change of base formula

$\ln ab = \ln a + \ln b$

i. $$\frac{d}{dx}\log_5 7x = \frac{d}{dx}[\log_5 7 + \log_5 x]$$
$$= \frac{d}{dx}\left[\log_5 7 + \frac{\ln x}{\ln 5}\right]$$
$$= \frac{1}{\ln 5}\cdot\frac{1}{x}$$

ii. $$\frac{d}{dx}\log_5 2020x = \frac{d}{dx}[\log_5 2020 + \log_5 x]$$
$$= \frac{d}{dx}\left[\log_5 2020 + \frac{\ln x}{\ln 5}\right]$$
$$= \frac{1}{\ln 5}\cdot\frac{1}{x}$$

iii. $$\frac{d}{dx}\log_5 369x = \frac{d}{dx}[\log_5 369 + \log_5 x]$$
$$= \frac{d}{dx}\left[\log_5 369 + \frac{\ln x}{\ln 5}\right]$$
$$= \frac{1}{\ln 5}\cdot\frac{1}{x}$$

Observation: All 3 expressions share the same derivatives. The constant factor in each expression becomes a constant term on its own because $\log_5 kx = \log_5 k + \log_5 x$, so its derivative is 0. The remaining term $\log_5 x$ is shared by all 3 expressions, so the final derivative is the same for all 3 expressions.

Question 2c

Let $f(x) = ae^{a-x}$, defined over the domain $-a \le x \le a$, where $a \in \mathbb{R}, a > 0$. Determine an expression for $f^{-1}(x)$, as well as its domain and range.

(7 marks)

$-a \le x \le a$

$-a \le -x \le a$

$a-a \le a-x \le a+a$

$0 \le a-x \le 2a$

$e^0 \le e^{a-x} \le e^{2a}$

$ae^0 \le ae^{a-x} \le ae^{2a}$

$\therefore a \le ae^{a-x} \le ae^{2a}$

range of $f(x) = [a, ae^{2a}]$

domain of $f^{-1}(x)$ = range of $f(x) = [a, ae^{2a}]$

range of $f^{-1}(x)$ = domain of $f(x) = [-a, a]$

• preliminary check: check if $f(x)$ is 1-to-1

if $f(x)$ is not 1-to-1, $f^{-1}(x)$ doesn't exist.

$\ln e^{a-x}$

let $y = ae^{a-x}$

$\ln y = \ln ae^{a-x} = \ln a + (a-x)$

$x = \ln a + a - \ln y$

$\therefore f^{-1}(x) = \ln a + a - \ln x$

Question 3a

Let $g : \mathbb{R} \to \mathbb{R}$ be a function such that

$$\int_0^1 (x-a)^2 g(x)dx = 0$$

where $a \in \mathbb{R}, a > 0$. If

$$\int_0^1 x^2 g(x)dx = a^2 \quad \text{and} \quad \int_0^1 g(x)dx = 3,$$

determine the value of

$$\int_0^1 xg(x)dx$$

in terms of a.

(6 marks)

Strategy: observe x^2, x are terms of $(x-a)^2$ upon expansion

$(x-a)^2 = x^2 - 2ax + a^2$

$(x-a)^2 g(x) = x^2 g(x) - 2a(x g(x)) + a^2 g(x)$

concept is $\int_a^b f(x) + g(x)\, dx = \int_a^b f(x)\, dx + \int_a^b g(x)\, dx$

$\int_0^1 (x-a)^2 g(x)\, dx$

$= \int_0^1 (x^2 - 2ax + a^2) g(x)\, dx$

$= \int_0^1 x^2 g(x)\, dx - 2a \int_0^1 x g(x)\, dx + a^2 \int_0^1 g(x)\, dx$

$= a^2 + 2a \int_0^1 x g(x)\, dx + 3a^2$

$= 4a^2 + 2a \int_0^1 x g(x)\, dx$

since $\int_0^1 (x-a)^2 g(x) = 0$,

$4a^2 + 2a \int_0^1 x g(x)\, dx = 0$

$2a \int_0^1 x g(x)\, dx = -4a^2$

$\therefore \int_0^1 x g(x)\, dx = -2a$

Question 3b

Interpret

$$\lim_{n\to\infty}\sum_{k=1}^{n}\frac{k^4}{(2n)^5}$$

as a definite integral and evaluate it.

(7 marks)

$$\lim_{n\to\infty}\sum_{k=1}^{n}\frac{k^4}{(2n)^5}$$

$$= \lim_{n\to\infty}\sum_{k=1}^{n}\frac{1}{2n}\cdot\left(\frac{k}{2n}\right)^4$$

$$= \lim_{n\to\infty}\sum_{k=1}^{n}\underbrace{\frac{\frac{1}{2}}{n}}_{\Delta x}\underbrace{\left[\frac{\frac{1}{2}}{n}k+0\right]}_{x_i}^4$$

$$= \int_0^{\frac{1}{2}} x^4\,dx$$

$$= \left[\frac{1}{5}x^5\right]_0^{\frac{1}{2}}$$

$$= \frac{1}{160}$$

consider $\int_a^b f(x)\,dx$,

$\Delta x = \frac{b-a}{n}$

$x_i = x_0 + i\Delta x$
$= a + i\Delta x$

interval $= [0, \frac{1}{2})$

Question 3c(i)

By using the trigonometric identity $\cos^2 x = 1 - \sin^2 x$, determine the indefinite integral

$$\int \cos^3 x\,dx$$

(4 marks)

strategy: observe odd power of cosine
$\Rightarrow \cos^{2k+1} x = \cos^{2k} x \cdot \cos x$

$$\int \cos^3 x\,dx = \int \cos^2 x \underbrace{\cos x\,dx}_{du = \cos x\,dx \Rightarrow u = \sin x}$$

$$= \int (1-\sin^2 x)\,du$$

$$= \int 1 - u^2\,du$$

$$= u - \frac{1}{3}u^3 + C$$

$$= \sin x - \frac{1}{3}\sin^3 x + C,$$ where C is a constant

Question 3c(ii)

Show that the use of the following reduction formula achieves the same answer as in Question 3(c)(i):

$$\int \cos^n x\,dx = \frac{1}{n}\cos^{n-1} x \sin x + \frac{n-1}{n}\int \cos^{n-2} x\,dx$$

(3 marks)

$$\int \cos^3 x\,dx = \frac{1}{3}\cos^2 x \sin x + \frac{3-1}{3}\int \cos x\,dx$$

$$= \frac{1}{3}\cos^2 x \sin x + \frac{2}{3}\sin x + C$$

$$= \frac{1}{3}\left[\cos^2 x \sin x + 2\sin x\right] + C$$

$$= \frac{1}{3}\left[(1-\sin^2 x)\sin x + 2\sin x\right] + C$$

$$= \frac{1}{3}\left[\sin x - \sin^3 x + 2\sin x\right] + C$$

$$= \frac{1}{3}\left[3\sin x - \sin^3 x\right] + C$$

$$= \sin x - \frac{1}{3}\sin^3 x + C$$

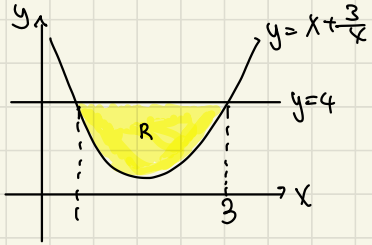
Question 4

Let R be the region in the first quadrant bounded by the curve $y = x + \frac{3}{x}$ and the line $y = 4$. [A sketch of the two curves might be helpful.]

Question 4a

Find the area of the region R.

(6 marks)



let $x + \frac{3}{x} = 4$

$x^2 + 3 = 4x$

$x^2 - 4x + 3 = 0$

$(x-3)(x-1) = 0$

$x = 1$ or 3

area of $R = \int_1^3 4 - (x + \frac{3}{x})\,dx$

$= \int_1^3 4 - x - \frac{3}{x}\,dx$

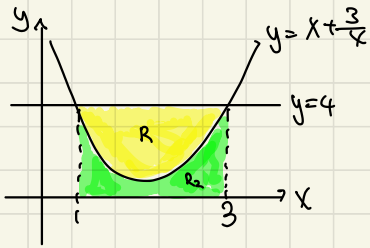
$= \left[4x - \frac{1}{2}x^2 - 3\ln|x|\right]_1^3$

$= 4\ln 3$

Question 4b

Using the disk/washer method, find the volume of the solid of revolution obtained when R is rotated 2π radians about the x-axis.

(6 marks)



consider solid of revolution for R_2

$r(x) = x + \frac{3}{x}$

volume of solid of revolution for R_2

$= \int_1^3 \pi [r(x)]^2 \, dx$

$= \int_1^3 \pi \left(x + \frac{3}{x}\right)^2 dx$

$= \pi \int_1^3 x^2 + 6 + \frac{9}{x^2} \, dx$

$= \pi \left[\frac{1}{3}x^3 + 6x - \frac{9}{x}\right]_1^3$

$= \frac{80\pi}{3}$

consider external cylinder.

i.e. solid of revolution when rotating $R + R_2$ about x-axis

$r(x) = 4$

volume of external cylinder

$= \int_1^3 \pi [r(x)]^2 \, dx$

$= \int_1^3 16\pi \, dx$

$= 32\pi$

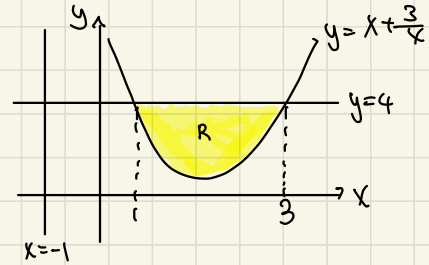
volume of solid of revolution for R

$= 32\pi - \frac{80}{3}\pi$

$= \frac{16}{3}\pi$

Question 4c

Using the method of cylindrical shells, find the volume of the solid of revolution obtained when R is rotated 2π radians about the line $x = -1$.



$r(x) = x + 1$

$h(x) = 4 - \left(x + \frac{3}{x}\right) = 4 - x - \frac{3}{x}$

$x > 0$

surface area of each shell $= 2\pi r(x) \cdot h(x)$

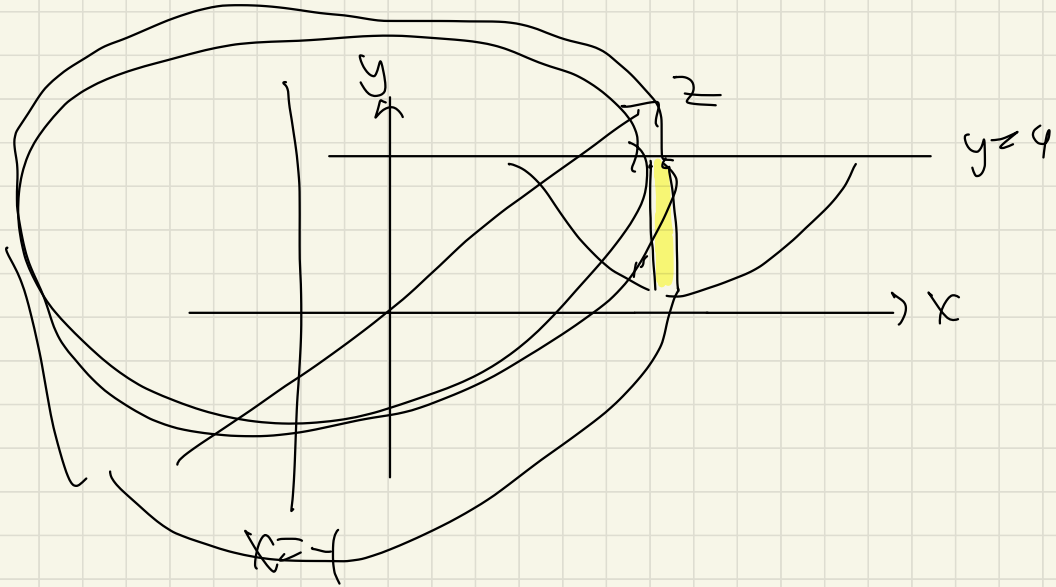
$= 2\pi (x+1)\left(4 - x - \frac{3}{x}\right)$

volume $= \int_1^3 2\pi (x+1)\left(4 - x - \frac{3}{x}\right) dx$

$= 2\pi \int_1^3 -x^2 + 3x + 1 - \frac{3}{x} \, dx$

$= 2\pi \left[-\frac{1}{3}x^3 + \frac{3}{2}x^2 + x - 3\ln|x|\right]_1^3$

$= 2\pi \left[\frac{16}{3} - 3\ln 3\right]$


y
z
y=4
x
x=-1

Question 5a

Determine the derivative of the following function and simplify it as much as possible:

$$y = \frac{1}{2}x\sin^{-1}\left(\frac{x}{2}\right) + \sqrt{1-\frac{x^2}{4}}\,.$$

(6 marks)

product rule chain rule

$$\frac{dy}{dx} = \frac{d}{dx}\left[\frac{x}{2}\sin^{-1}\frac{x}{2} + \sqrt{1-\frac{x^2}{4}}\right]$$

$$= \frac{1}{2}\sin^{-1}\frac{x}{2} + \frac{x}{2}\frac{\frac{1}{2}}{\sqrt{1-\frac{x^2}{4}}} + \frac{1}{2}\left(1-\frac{x^2}{4}\right)^{-\frac{1}{2}}\left(-\frac{x}{2}\right)$$

$$= \frac{1}{2}\sin^{-1}\frac{x}{2} + \frac{x}{2}\frac{\frac{1}{2}}{\sqrt{1-\frac{x^2}{4}}} - \frac{x}{2}\frac{\frac{1}{2}}{\sqrt{1-\frac{x^2}{4}}}$$

$$= \frac{1}{2}\sin^{-1}\frac{x}{2}$$

cancel out

Question 5b

Let $f : \mathbb{R} \to \mathbb{R}$ be a continuous function. Solve for

$$\frac{d}{dx}\int_0^x t f(x^2 - t^2)\,dt\,,$$

giving your answer in terms of x and f.

(6 marks)

strategy: convert expression to a familiar form

let $u = x^2 - t^2 \Rightarrow du = -2t\,dt$,

when $u=0$, $x^2 = t^2 \Rightarrow t = x$

when $t=0$, $u = x^2$

$$\int_0^x t f(x^2-t^2)\,dt = -\frac{1}{2}\int_0^x f(x^2-t^2)(-2t)\,dt$$

$$= -\frac{1}{2}\int_{x^2}^0 f(u)\,du$$

$$= \frac{1}{2}\int_0^{x^2} f(u)\,du$$

By FTC,

$$\int_0^x t f(x^2-t^2)\,dt = \frac{1}{2}\int_0^{x^2} f(u)\,du = \frac{1}{2}\left[F(x^2) - F(0)\right]$$

$$\frac{d}{dx}\int_0^x t f(x^2-t^2)\,dt = \frac{d}{dx}\,\frac{1}{2}\left[F(x^2) - F(0)\right]$$

$$= \frac{1}{2}\left[2x f(x^2) - 0\right]$$

$$= x f(x^2)$$

Question 5c

By using the substitution $x = \frac{\pi}{4} - t$ or otherwise, determine the exact value of

$$\int_0^{\frac{\pi}{4}} 2\ln\sqrt{1+\tan x}\,dx\,.$$

(8 marks)

$$1+\tan x = 1+\tan\left(\frac{\pi}{4}-t\right)$$

$$= 1+\frac{\tan\frac{\pi}{4}-\tan t}{1+\tan\frac{\pi}{4}\tan t} \quad \Big\}\ \tan(A-B) = \frac{\tan A-\tan B}{1-\tan A\tan B}$$

$$= 1+\frac{1-\tan t}{1+\tan t}$$

$$= \frac{2}{1+\tan t} \quad \Big\}\ \text{observe denominator \& LHS same expression}$$

$x = \frac{\pi}{4}-t \quad dx = -dt,$ when $x=0,\ t=\frac{\pi}{4}$

when $x=\frac{\pi}{4}\quad t=0$

$$\int_0^{\frac{\pi}{4}} 2\ln\sqrt{1+\tan x}\,dx = \int_{\frac{\pi}{4}}^{0} 2\ln\sqrt{\frac{2}{1+\tan t}}\,(-1)\,dt$$

$$= \int_0^{\frac{\pi}{4}} 2\ln\sqrt{\frac{2}{1+\tan t}}\,dt \qquad \ln\frac{a}{b} = \ln a - \ln b$$

$$= \int_0^{\frac{\pi}{4}} 2\ln\sqrt{2}\,dt - \underbrace{\int_0^{\frac{\pi}{4}} 2\ln\sqrt{1+\tan t}\,dt}_{\text{generate term from LHS}}$$

observe $\int_0^{\frac{\pi}{4}} 2\ln\sqrt{1+\tan x}\,dx = \int_0^{\frac{\pi}{4}} 2\ln\sqrt{1+\tan t}\,dt$

$$\Rightarrow 2\int_0^{\frac{\pi}{4}} 2\ln\sqrt{1+\tan x}\,dx = \int_0^{\frac{\pi}{4}} 2\ln\sqrt{2}\,dt$$

$$= \left(2\ln\sqrt{2}\right)\frac{\pi}{4}$$

$$\therefore \int_0^{\frac{\pi}{4}} 2\ln\sqrt{1+\tan x}\,dx = \frac{\pi}{4}\ln\sqrt{2}$$

$$= \frac{\pi}{4}\left(\frac{1}{2}\right)\ln 2$$

$$= \frac{\pi}{8}\ln 2$$